PRATHYUSHA TALLAPUREDDY

tprathyusha0559@gmail.com | (940) 364-0559

Senior Data Engineer | Python | PySpark | SQL | Snowflake | dbt | AWS | Azure Databricks | Airflow | Kafka

PROFESSIONAL SUMMARY

Senior Data Engineer with 5+ years of experience designing and operating ETL/ELT pipelines, warehouses, and lakehouses for healthcare and financial services. Processes 5–10 million records daily with 99.2% SLA compliance and 30% faster pipeline runtime. Expertise in Python, PySpark, SQL, dbt, Airflow, Kafka, AWS, Snowflake, and Azure Databricks; dimensional modeling, Delta Lake Medallion architecture, automated quality, and HIPAA-aligned governance. Databricks and Snowflake certified, with CI/CD, Power BI, Agile collaboration, and GenAI experience using RAG and Amazon Bedrock.

TECHNICAL SKILLS

Core Competencies: Data Pipelines, ETL (Extract, Transform, Load) / ELT (Extract, Load, Transform), Data Warehousing, Data Integration, Data Lakehouse, Data Modeling, Workflow Orchestration, Real-Time Streaming, Data Quality, Data Governance, SLA Management, Cross-Functional Collaboration

Languages: Python, PySpark, SQL, pandas, NumPy, Shell Scripting

Pipelines & Orchestration: Apache Airflow, dbt, AWS Glue, Azure Data Factory, Change Data Capture (CDC), Incremental Loads, Batch and Streaming ETL/ELT

Big Data & Lakehouse: Apache Spark, Spark Structured Streaming, Delta Lake, Apache Kafka, Medallion Architecture (Bronze/Silver/Gold)

AWS: S3, Glue, Redshift, Athena, CloudWatch, IAM, Amazon Bedrock

Azure: Azure Databricks, ADLS Gen2, Data Factory, Synapse Analytics, Azure SQL

Warehousing & Data Modeling: Snowflake, Redshift, Synapse, Dimensional Modeling (Kimball), Star Schema, Fact/Dimension Tables

Data Quality: Great Expectations, dbt Tests, Data Profiling, Schema Validation, Reconciliation

Databases: PostgreSQL, Azure SQL, Query Optimization

BI: Power BI, Excel

DevOps & CI/CD: CI/CD (Continuous Integration/Continuous Delivery), Git, GitHub Actions, Jenkins, Docker, Azure DevOps

Governance & Compliance: HIPAA, PHI (Protected Health Information) Handling, Data Governance, Access Controls, Audit Logging

Generative AI: LLMs, Retrieval-Augmented Generation (RAG), Amazon Bedrock, Anthropic Claude

Practices: Agile/Scrum, Performance Tuning, Pipeline Monitoring, Technical Documentation

PROFESSIONAL EXPERIENCE

Blue Cross Blue Shield Dallas, TX

Senior Data Engineer *Aug 2025 – Present*

- Achieved 99.2% SLA compliance (up from 88%) on daily pharmacy data refreshes by orchestrating 40+ Airflow DAGs for AWS Glue, PySpark, and Snowflake dbt runs, cutting pipeline processing time by 30% through partition tuning and incremental materialization.
- Optimized Snowflake SQL queries, clustering keys, and dbt models for clinical-quality reporting, reducing warehouse compute costs by 25%.
- Built and operated production Python and PySpark ETL pipelines processing 5–10 million pharmacy, adherence, formulary, and clinical-quality PHI records daily in AWS S3 and Snowflake with HIPAA-aligned controls and Delta Lake Bronze, Silver, and Gold layers.
- Designed modular Snowflake dbt models to separate pharmacy business logic into reusable transformations for adherence, formulary utilization, drug costs, and clinical quality, with referential-integrity and business-rule tests, alongside Athena queries on curated S3 datasets.
- Automated schema, completeness, NDC-format, formulary-tier, and business-rule checks with Great Expectations and dbt, reducing validation errors and manual QA effort.
- Strengthened PHI access controls with field-level masking, Snowflake column-level access controls, and audit logging for member data.
- Built Kafka consumers that validated pharmacy transactions and formulary changes before loading events into Snowflake for near-real-time utilization reporting.
- Built a RAG workflow with cited pharmacy policies for review; used Bedrock and Claude for documentation and SQL checks.
- Packaged Python jobs with Docker; automated pipeline tests and deployments with GitHub Actions.
- Configured Airflow dependency checks and alerts to detect delayed refreshes and support pipeline recovery.

Environment: Python, PySpark, SQL, Snowflake, dbt, AWS S3, Glue, Athena, Delta Lake, Airflow, Kafka, Great Expectations, Docker, GitHub Actions, Bedrock, Claude, RAG, HIPAA/PHI Controls.

American Express Hyderabad, India
Senior Data Engineer *Jul 2022 – Aug 2024*

- Built Spark Structured Streaming pipelines with Kafka on Azure Databricks, ingesting 1.2M+ card transaction events per hour into Delta Lake (Bronze/Silver) with sub-minute latency for near-real-time rewards and fraud analytics.
- Reduced daily Databricks cluster execution time by 25% through Spark repartitioning, liquid clustering, and query tuning, lowering monthly Azure compute costs.
- Built Python and PySpark pipelines on Azure Databricks to ingest card transactions, rewards redemptions, and member spend into ADLS, transforming and publishing clean datasets across Delta Lake Bronze, Silver, and Gold layers.
- Automated incremental loads with parameterized Azure Data Factory and Databricks workflows into Synapse, Azure SQL, and PostgreSQL, deploying changes through Azure DevOps CI/CD.
- Applied SQL window functions and PySpark null, duplicate, and schema checks before publishing curated datasets to downstream reporting.
- Translated product and analytics requirements into pipeline designs in Agile sprints, delivering Power BI rewards and merchant dashboards on Synapse.
- Containerized Python jobs with Docker and used GitHub Actions to automate tests and pipeline deployments.
- Aligned curated transaction and rewards datasets with business reporting requirements, applying consistent validation before publication.

Environment: Python, PySpark, SQL, Azure Databricks, Structured Streaming, ADLS Gen2, ADF, Synapse, Azure SQL, PostgreSQL, Delta Lake, Kafka, Power BI, Azure DevOps, Docker, GitHub Actions, CI/CD.

BHG Financial Hyderabad, India
Data Engineer *Jul 2020 – Jun 2022*

- Redesigned Redshift dimensional schemas (Kimball methodology) and tuned distribution and sort keys across 15+ fact and dimension tables, improving portfolio query execution times by 40%.
- Built Airflow DAGs to orchestrate daily AWS Glue extractions and Redshift loads, consistently meeting underwriting SLAs with 99.5% pipeline uptime and delivering underwriting and portfolio risk reports 2 hours earlier each morning.
- Built production Python and PySpark ETL pipelines that extracted loan origination, credit risk, and borrower data from PostgreSQL source databases into AWS S3 and Redshift.
- Automated loan-count and balance reconciliation in Python between source PostgreSQL databases and the data warehouse, improving stakeholder trust in lending and portfolio reports.
- Staged raw and curated loan data in partitioned AWS S3 buckets with Glue Data Catalog tables for validation, efficient querying, and reliable Redshift ingestion.
- Built Power BI dashboards for portfolio and credit-risk reporting while maintaining Jenkins CI/CD pipelines for ETL deployments and documenting pipeline logic for audits.
- Documented source-to-target design decisions for loan and borrower datasets to support reconciliation, audit traceability, and team handoff.

Environment: Python, PySpark, SQL, PostgreSQL, AWS S3, AWS Glue, Glue Crawlers, Redshift, Airflow, Power BI, Jenkins, CI/CD.

CERTIFICATIONS

- Databricks Certified Data Engineer Professional
- Snowflake SnowPro Core Certification
- Anthropic AI Fluency: Framework & Foundations

EDUCATION

University of North Texas Denton, TX
Master of Science, Advanced Data Analytics | GPA: 4.0 *Aug 2024 – May 2026*